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Golden Dome for America

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Israeli Multi-Layered Missile Defense Architecture



Iron Dome
Innermost Layer

Tamir interceptors engage
threats <70km away
Rafael w/ US funding



David's Sling
Intermediate Layer

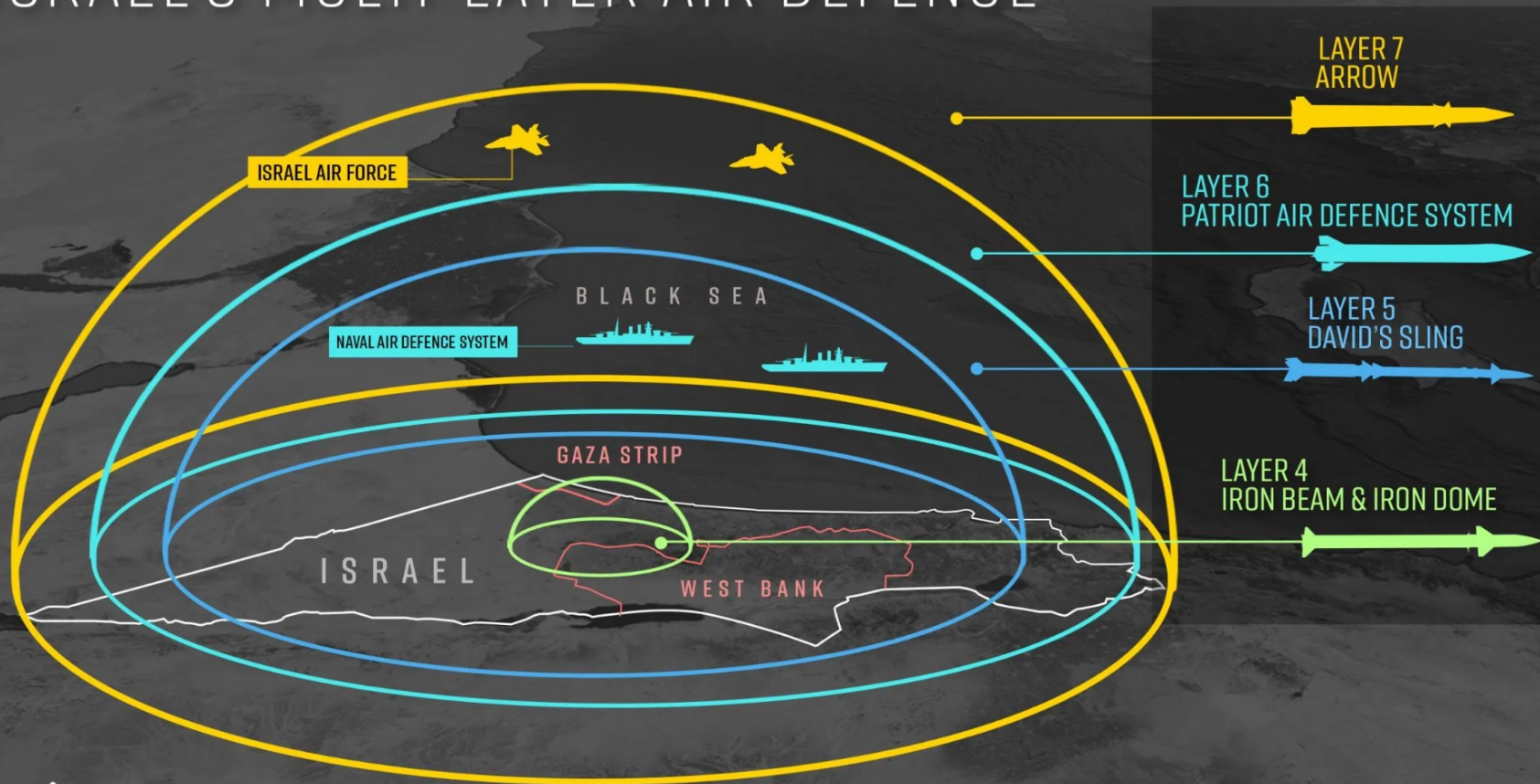
Stunner interceptors have a
range up to 300km, hit-to-kill
Rafael & Raytheon



Arrow
Top Layer

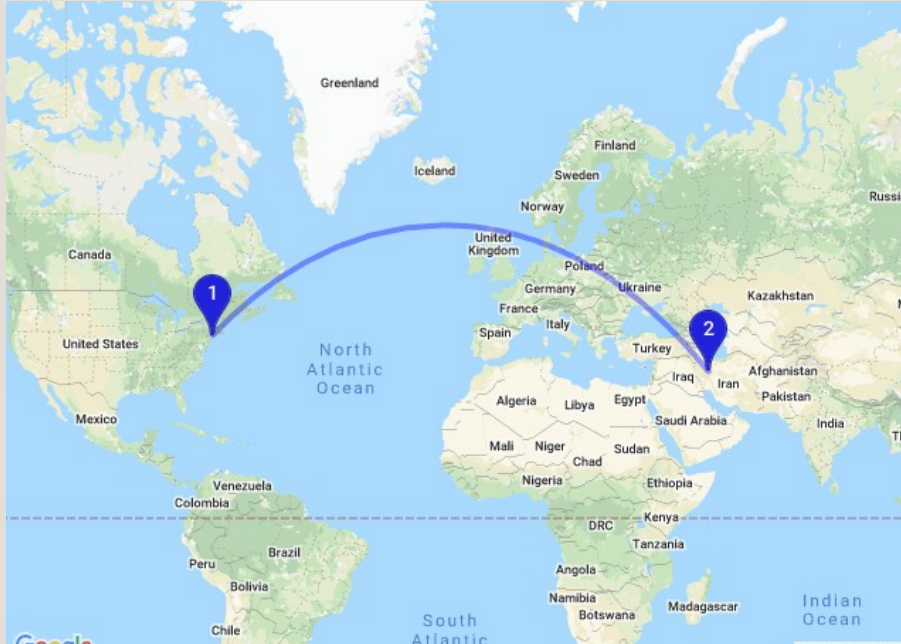
High altitude & long-range
ballistic missiles
Arrow 3 has a 100km+ max
altitude and 2400km range,
hit-to-kill
IAI & Boeing

ISRAEL'S MULTI-LAYER AIR DEFENCE



US to Adversaries

2. America's Differing Geopolitical Context



10,000km NYC to Tehran
~4.2x farther than Arrow 3 range



9,700km LA to Beijing
~4x farther than Arrow 3 range

E.O. 14186

Sections

1

Raison D'être

“The threat of attack by ballistic, hypersonic, and cruise missiles, and other advanced aerial attacks, remains the most catastrophic threat facing the United States.”

2

Policy

(a) “The United States will provide for the common defense of its citizens and the Nation by deploying and maintaining a next-generation missile defense shield”

(b) “The United States will deter — and defend its citizens and critical infrastructure against — any foreign aerial attack on the Homeland”

(c) “The United States will guarantee its secure second-strike capability”

3

Implementation

Next Slide



4

Allies

(a) “Increase bilateral and multilateral cooperation on missile defense technology development, capabilities, and operations”

(b) “Improve theater missile defenses of forward-deployed United States troops and allied territories, troops, and populations”

(c) “Increase and accelerate the provision of United States missile defense capabilities to allies and partners”

3. The Golden Dome Proposal

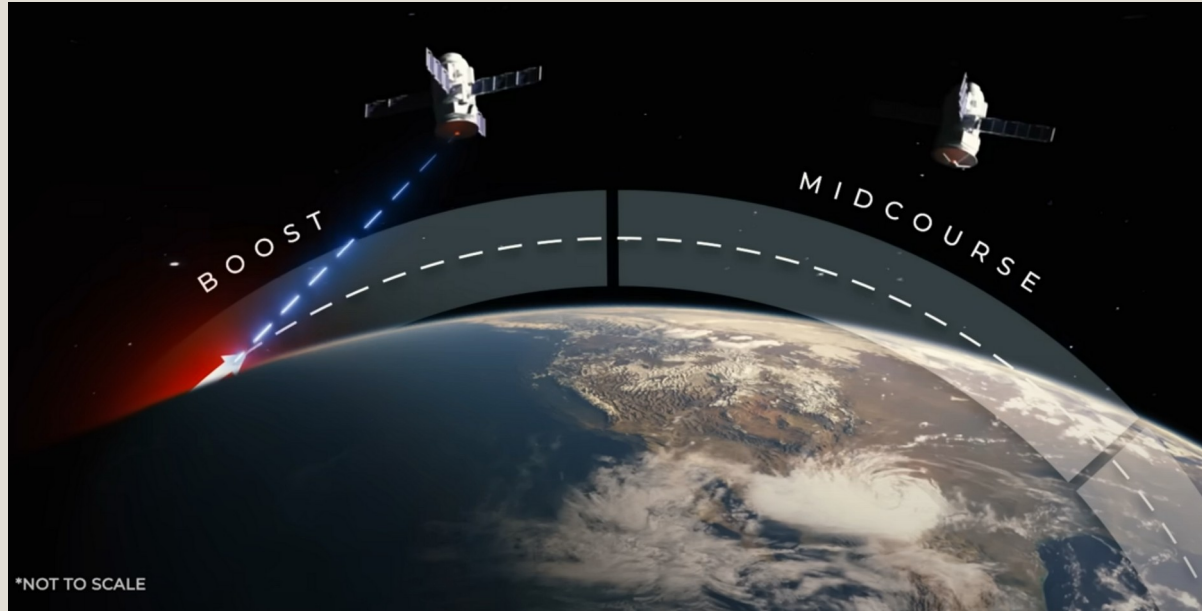
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Section 3

Implementation

- (i) Defense of the United States against ballistic, hypersonic, advanced cruise missiles, and other next-generation aerial attacks from peer, near-peer, and rogue adversaries
- (ii) Acceleration of the deployment of the Hypersonic and Ballistic Tracking Space Sensor layer
- (iii) Development and deployment of proliferated space-based interceptors capable of boost-phase intercept**
- (iv) Deployment of underlayer and terminal-phase intercept capabilities postured to defeat a countervalue attack**
- (v) Development and deployment of a custody layer of the Proliferated Warfighter Space Architecture
- (vi) Development and deployment of capabilities to defeat missile attacks prior to launch and in the boost phase
- (vii) Development and deployment of a secure supply chain for all components with next-generation security and resilience features
- (viii) Development and deployment of non-kinetic capabilities to augment the kinetic defeat of ballistic, hypersonic, advanced cruise missiles, and other next-generation aerial attacks**

Boost Phase Intercept



Why boost phase?

- Missile is still slow
- Bright IR signature
- Hasn't deployed decoys or multiple warheads yet

3.(a).(iii) “Development and deployment of proliferated **space-based** interceptors capable of **boost-phase intercept**”

Estimates range from 1,300-2000 satellites in LEO to provide truly global coverage

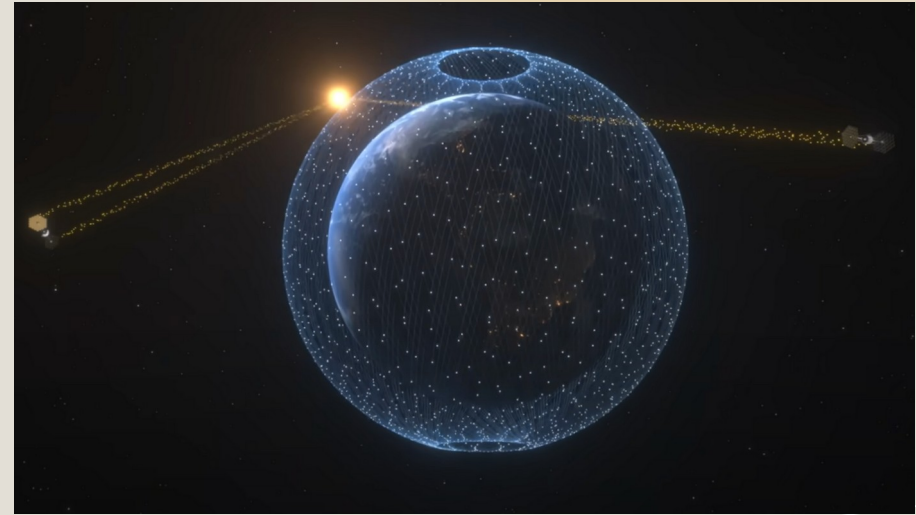
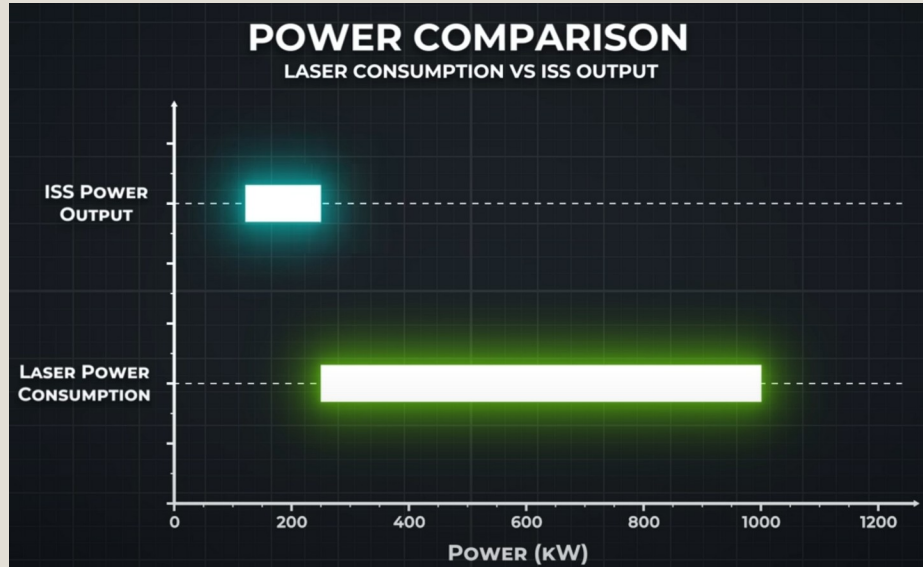
(Starlink has ~7000 satellites)

Layered Architecture



3.(a).(iv) “Deployment of underlayer and **terminal-phase intercept** capabilities postured to defeat a countervalue attack”

Directed Energy Weapons (DEW)

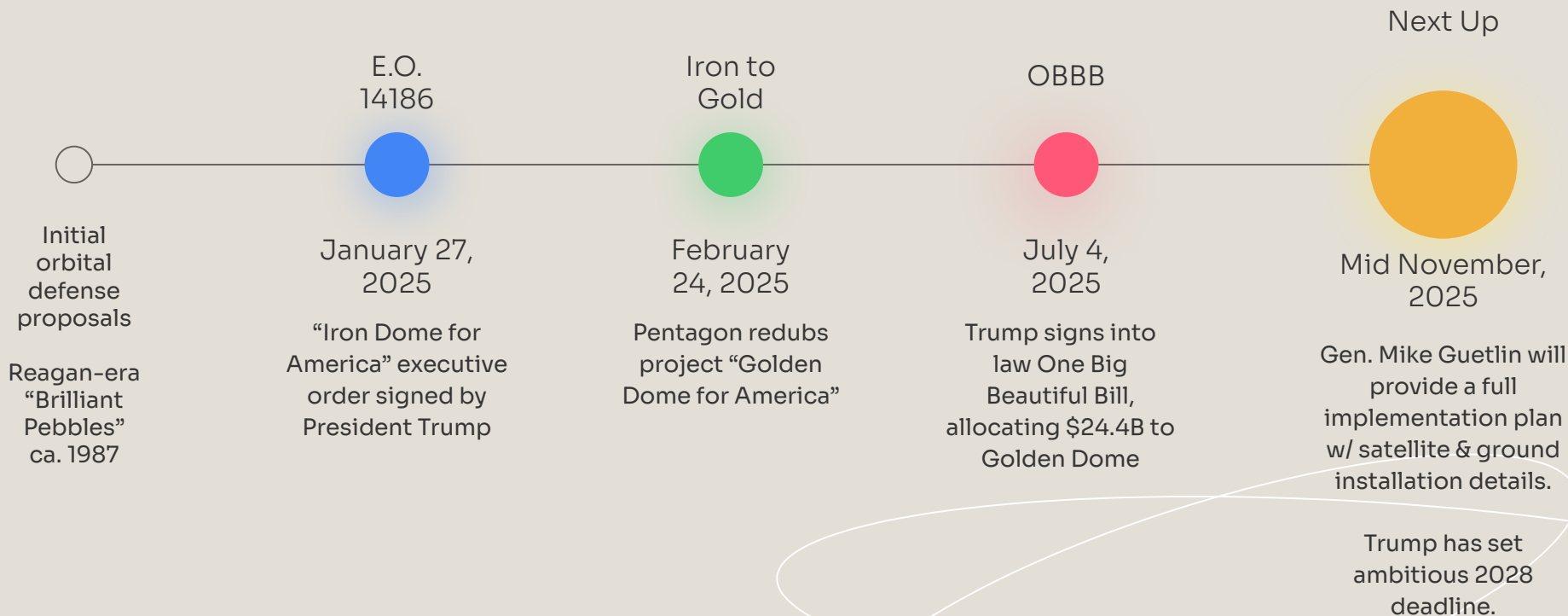


Estimates range from 1,300-2000 satellites in LEO to provide truly global coverage

(Starlink has ~7000 satellites)

3.(a).(viii) “Development and deployment of **non-kinetic capabilities** to augment the kinetic defeat of ballistic, hypersonic, advanced cruise missiles, and other next-generation aerial attacks”

Golden Dome Timeline



Domestic Concerns

Cost

Cost Israel \$285mn/day to operate Arrow system during Iran-Israel War in June 2025.

Trump: **\$175bn** / 3 years
Congressional Budget Office:
\$161bn-**\$252bn** / 20 yrs for just space-based interceptors

American Enterprise Institute
(DC Think Tank): \$252bn-**\$3.6tn**
final capability dependent

Gen. Guetlein: **>\$500bn**

Politicization

Given current partisan fight on the Hill over FY26 discretionary funds, a long-term project with a price tag like this one like this is subject to the Hill

- 2026 Midterms
- 2028 Presidential Race



Impacts on Space Tech & Ops

Mass Production & Launch

Ability to mass-produce & rapidly deploy satellites & interceptors into orbit, relying on falling launch costs and shortening time-to-orbit supported by private industry such as SpaceX's Falcon 9

C3I (Command, Control, Communications, Intelligence)

High speed, resilient, integrated military-grade network coordinating sensors, launchers, and command inputs across various domains (space, air, ground, sea) and various military branches (Space Force, Air Force, Navy)

Orbital Debris

While VLEO is a self-cleaning domain due to atmospheric drag, debris from tests or live combat skirmishes could render certain orbital regimes unusable

Impacts on International Affairs

Space Militarization

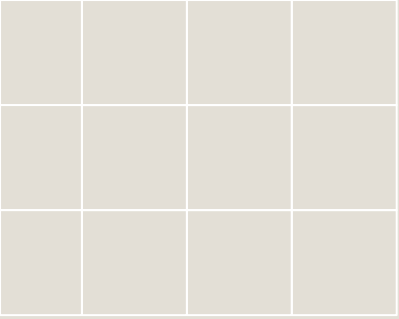
Viewed by US adversaries, namely China & Russia, as a destabilizing endeavor

Most cost-effective & immediate countermeasure is not constructing a comparable defense but increasing sheer volume & complexity of offensive weapons

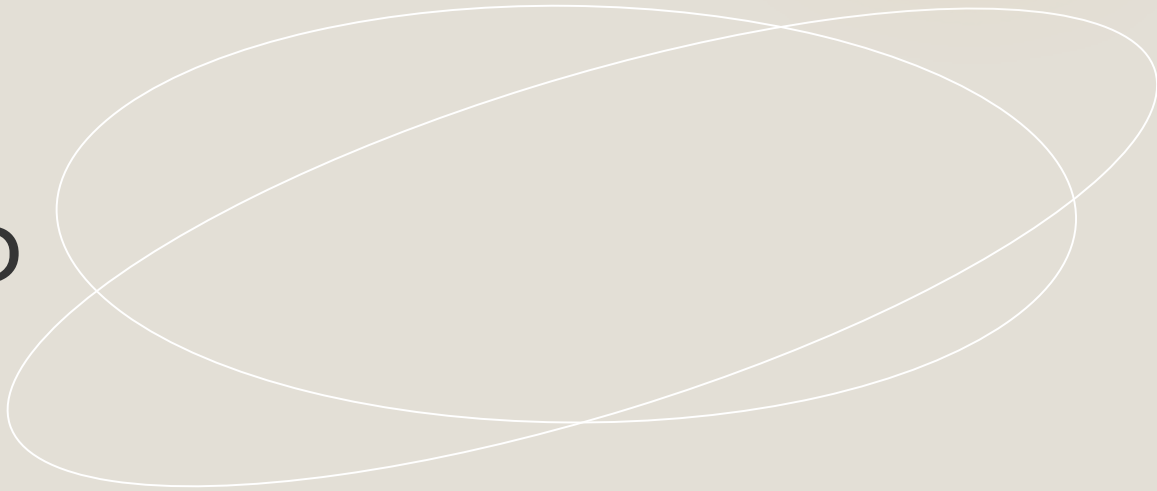
Outer Space Treaty (1967)

Golden Dome's kinetic interceptors are **not** prohibited by Article IV, which only prohibits nuclear weapons and other WMDs

Other proposed treaties such as PAROS & PPWT **would** ban such a system but have thus far **not** been adopted as binding international treaties



Questions?



Current US Missile Defense Architecture

THAAD

High-altitude
(40km-150km) endo &
exo-atmospheric
interception for
ballistic missiles, hit-
to-kill

PATRIOT

<40km altitude
tactical ballistic
missiles, cruise
missiles, aircraft, hit-
to-kill & blast
fragmentation

F-35

Airborne platform
using its phased array
antenna for earlier
tracking &
interception, air-to-air
missiles for boost
phase or cruise missile
flight